



Genômica Computacional

Anotação de genômas

Professor: Ricardo A. Vialle

CS31 - Genômica Computacional [27/07-30/07 - 12h00-14h00]

29 de Julho de 2026

Genome assembled, what next?

Adding biological info to sequences

ribosome
binding site

delta toxin
PubMed: 15353161

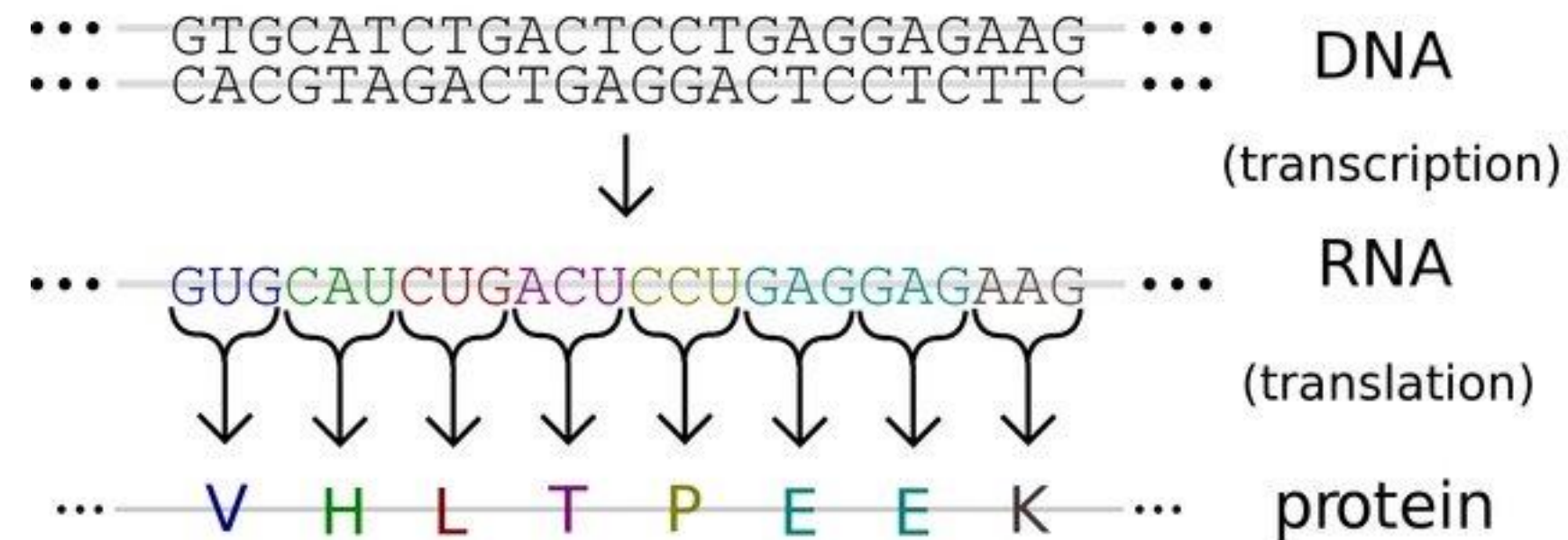
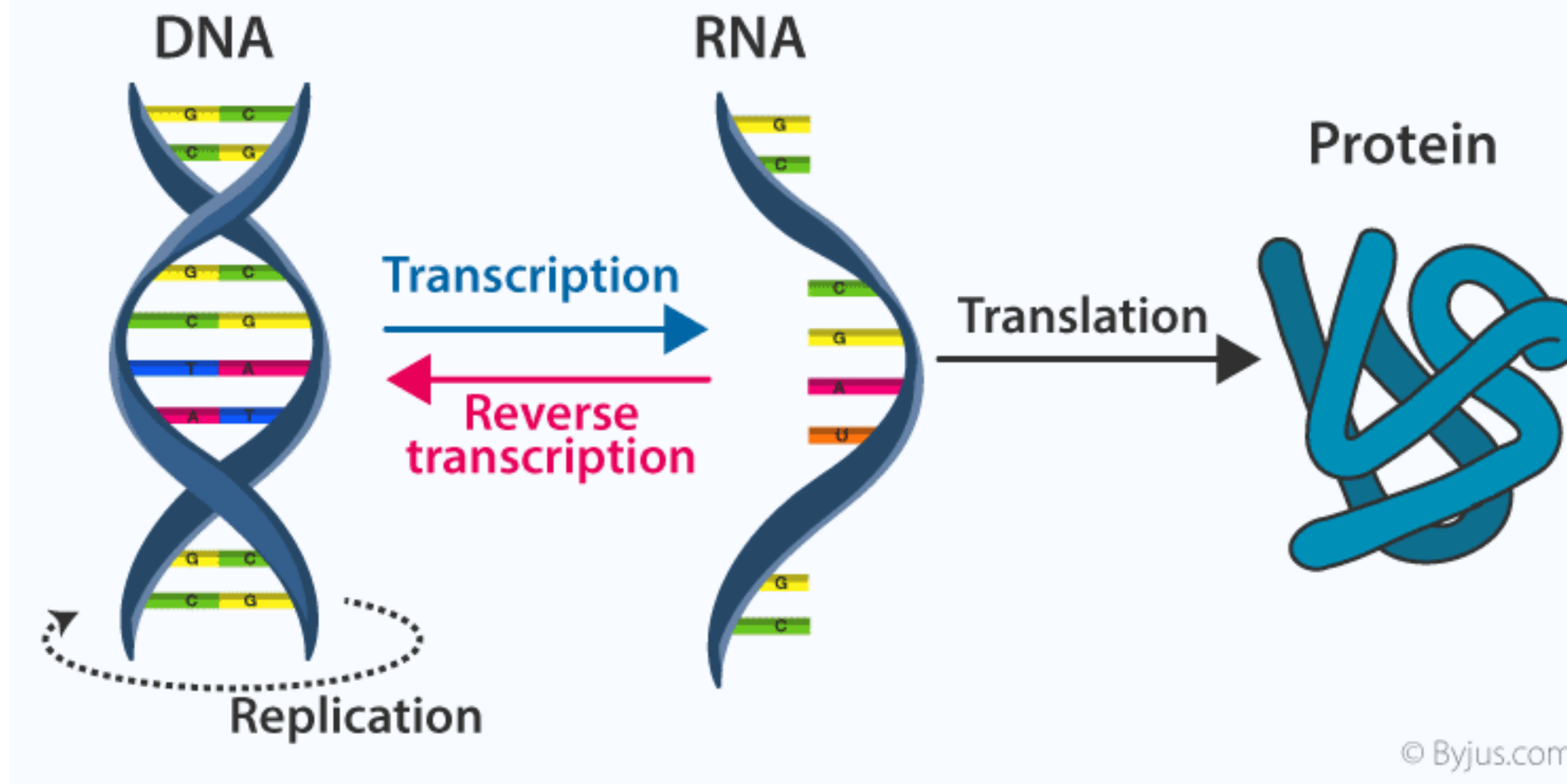
ACCGGCCGAGACAGCGAGCATATGCAGGAAGCGGCAGGAATAAGGA
AAAGCAGCCTCCTGACTTTCCTCGCTTGGTGGTTTGAGTGGACCTC
CCAGGCCAGTGCCGGGCCCCCTCATAGGAGAGGAAGCTCGGGAGGTG
GCCAGGCGGCAGGAAGGCGCACCCCCCAGCAATCCGCGCGCCGGG
ACAGAATGCCCTGCAGGAACCTTCTTCTAGAAGACCTTCTCCTCCTG
CAAATAAAACCTCACCCATGAATGCTCACGCAAGTTTAATTACAGA
CCTGAAACAAGATGCCATTGTCCCCCGGCCTCCTGCTGCTGCTGCT
CTCCGTCCGTCCGTGGGCCACGGCCACCGCTTTTTTTTTTTGCC

tandem repeat
CCGT x 3

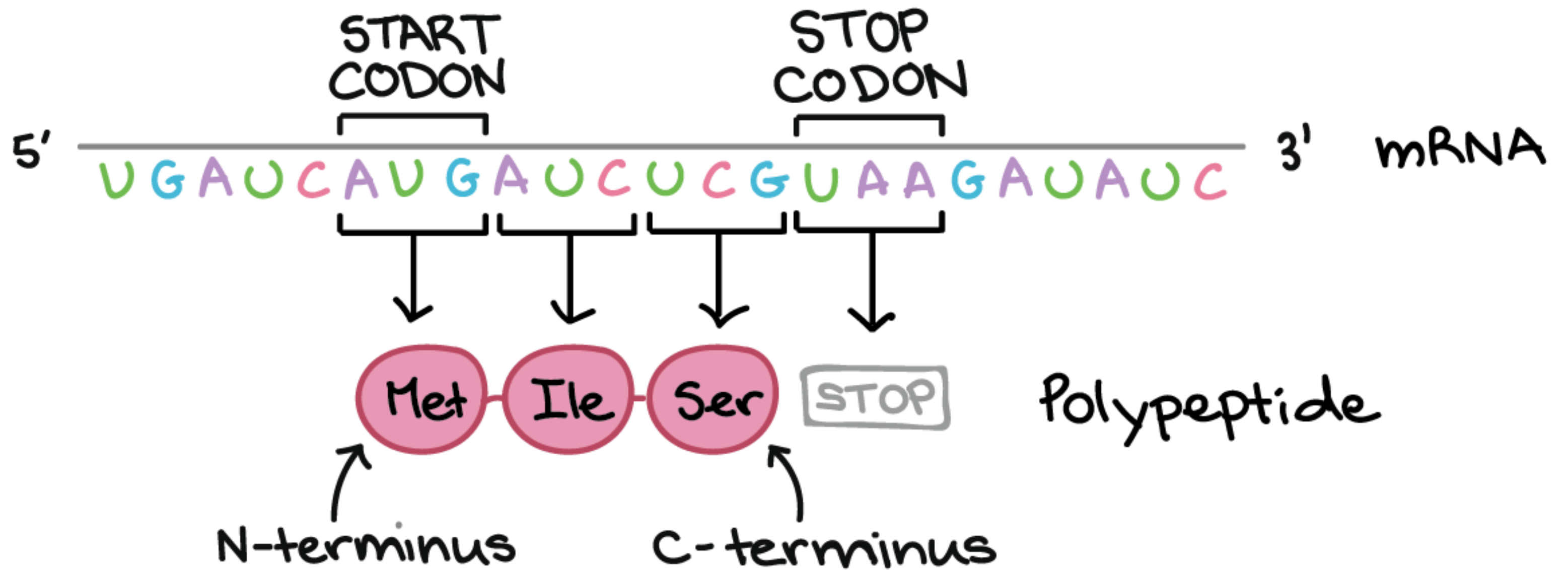
transfer RNA
Leu-(UUR)

homopolymer
10 x T

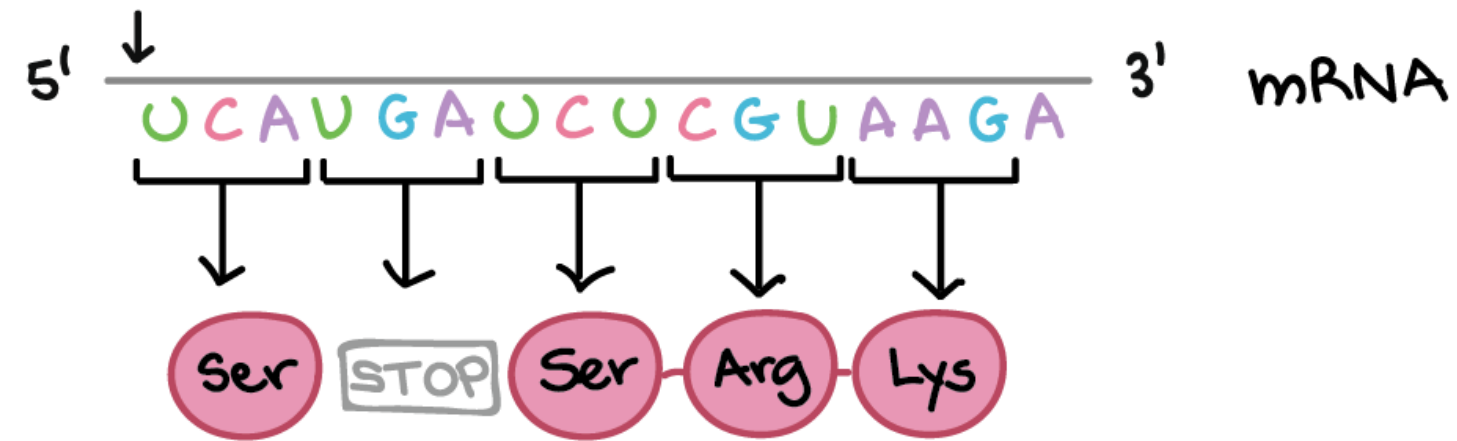
CENTRAL DOGMA : DNA TO RNA TO PROTEIN



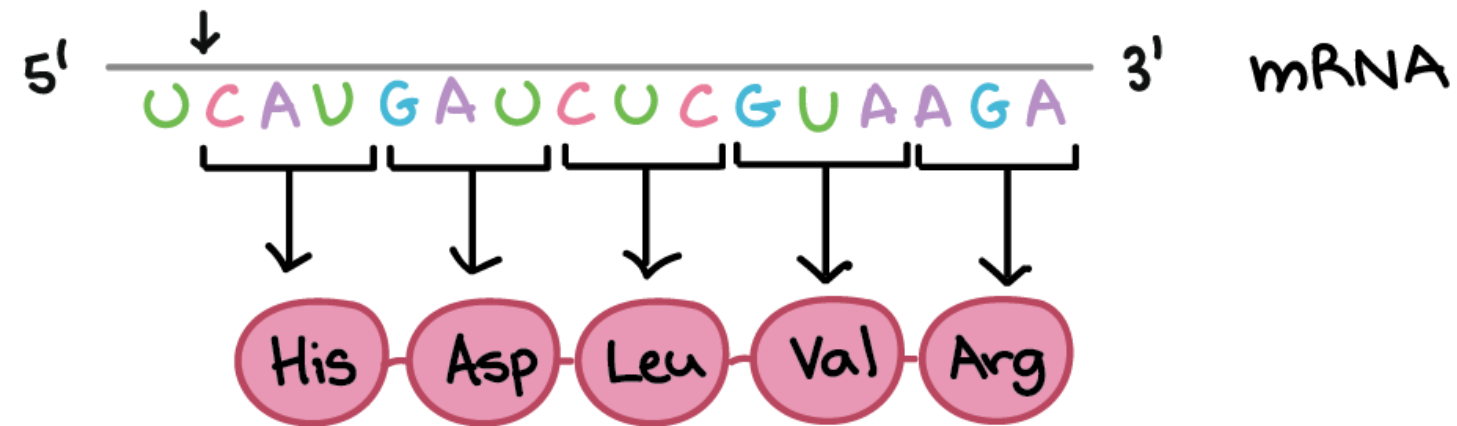
Segunda base						
Primera base	U	U	C	A	G	Tercera base
		UUU } Fenilalanina (Fen) UUC }	UCU } UCC } Serina (Ser) UCA } UCG }	UAU } Tirosina (Tir) UAC }	UGU } Cisteína (Cis) UGC }	
		UUA } Leucina (Leu) UUG }		UAA } Codón de terminación UAG }	UGA } Codón de terminación UGG } Triptófano (Tri)	
	C	CUU } CUC } Leucina (Leu) CUA } CUG }	CCU } CCC } Prolina (Pro) CCA } CCG }	CAU } Histidina (His) CAC } CAA } Glutamina (Gln) CAG }	CGU } CGC } Arginina (Arg) CGA } CGG }	
	A	AUU } AUC } Isoleucina (Ileu) AUA }	ACU } ACC } Treonina (Tre) ACA } ACG }	AAU } Asparagina (Asn) AAC } AAA } Lisina (Lis) AAG }	AGU } Serina (Ser) AGC } AGA } Arginina (Arg) AGG }	
	G	GUU } GUC } Valina (Val) GUA } GUG }	GCU } GCC } Alanina (Ala) GCA } GCG }	GAU } Ácido aspártico (Asp) GAC } GAA } Ácido glutámico (Glu) GAG }	GGU } GGC } Glicina (Gli) GGA } GGG }	



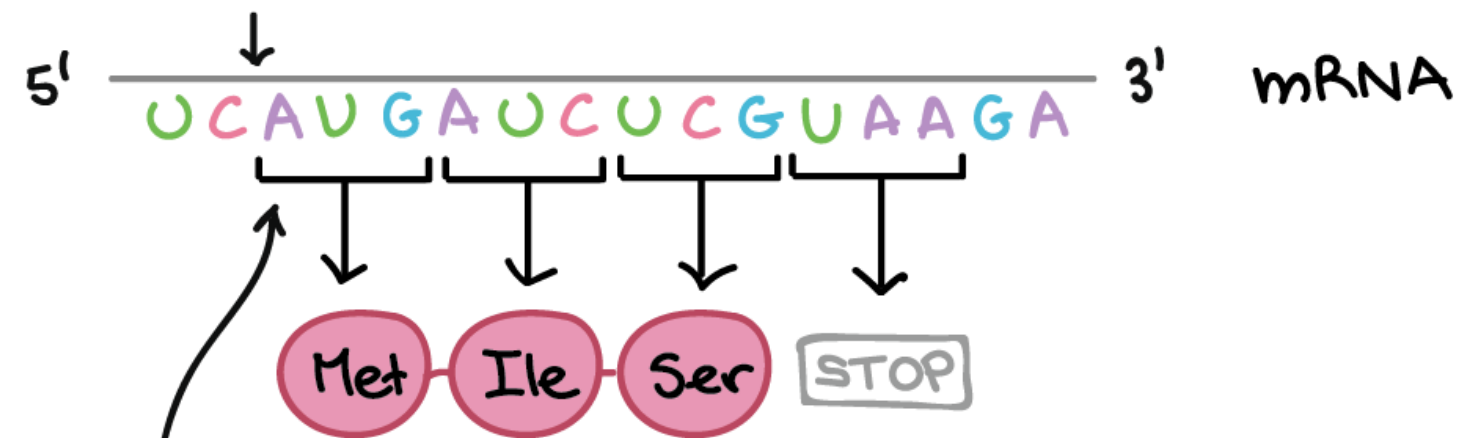
FRAME 1



FRAME 2

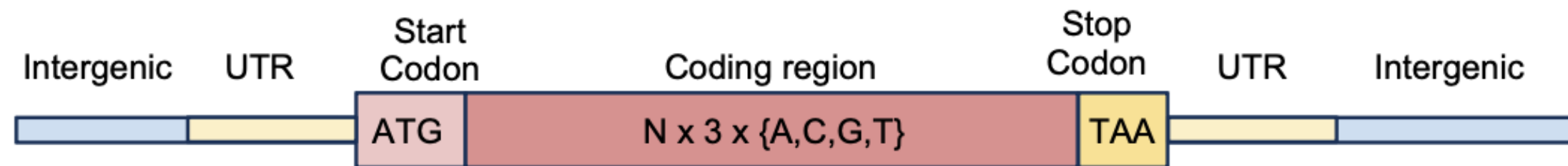


FRAME 3

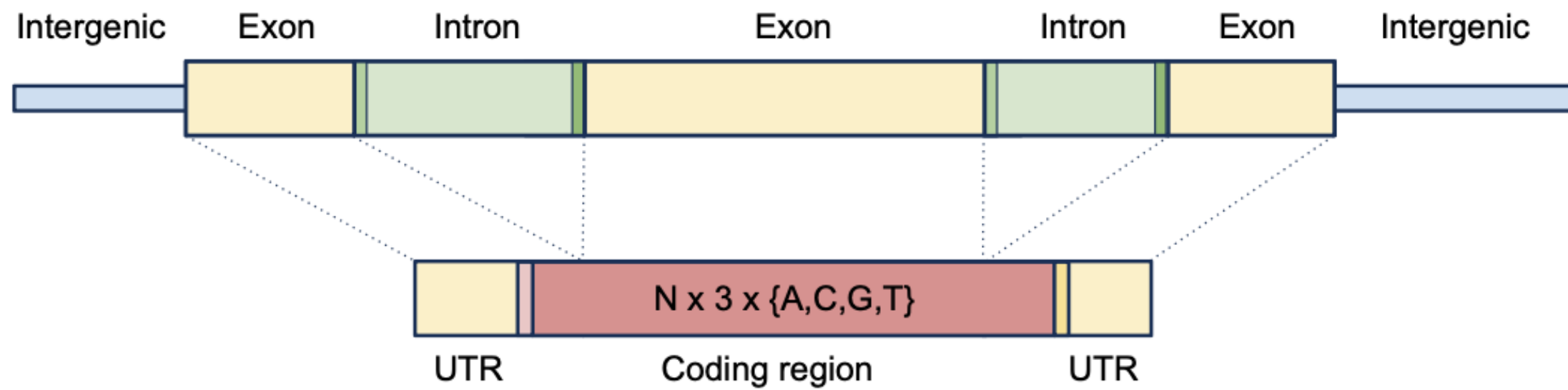


Start codon's position ensures that this frame is chosen

A) Prokaryotic Gene



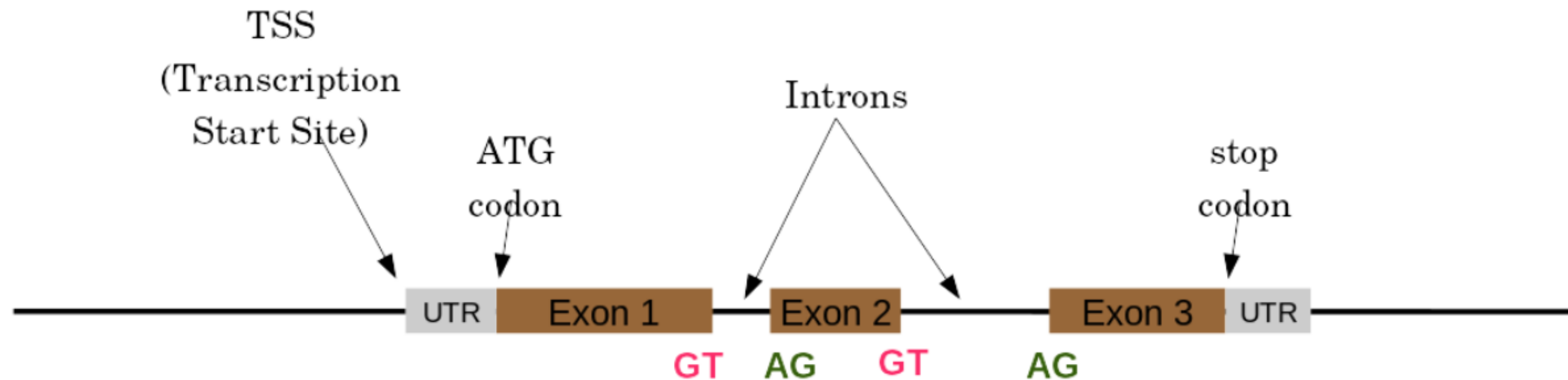
B) Eukaryotic Gene



Automatic Structural Annotation

Very difficult problem

- Short, variable, unspecific motifs
- Need data to support predictions



Strategies for identifying coding genes - Structural annotation

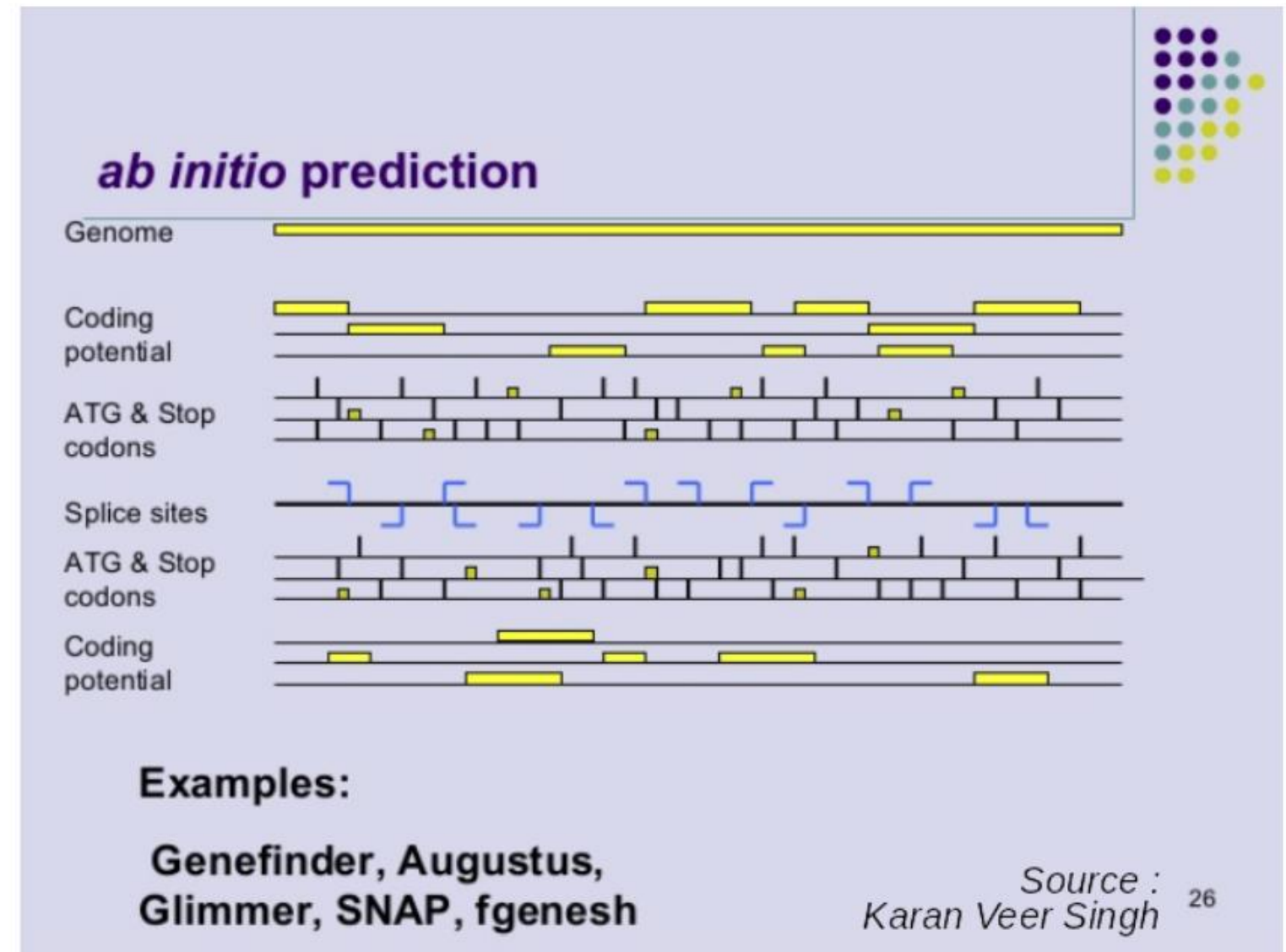
Ab initio Gene Calling

Predictions using:

- Genome sequence
- Statistical model (specific to organism)

Models:

- Training on the best evidence-based gene calls
- "Best" = strong evidence, highly conserved
- Training can be iterative:
 - train, predict, select best genes, retrain, etc

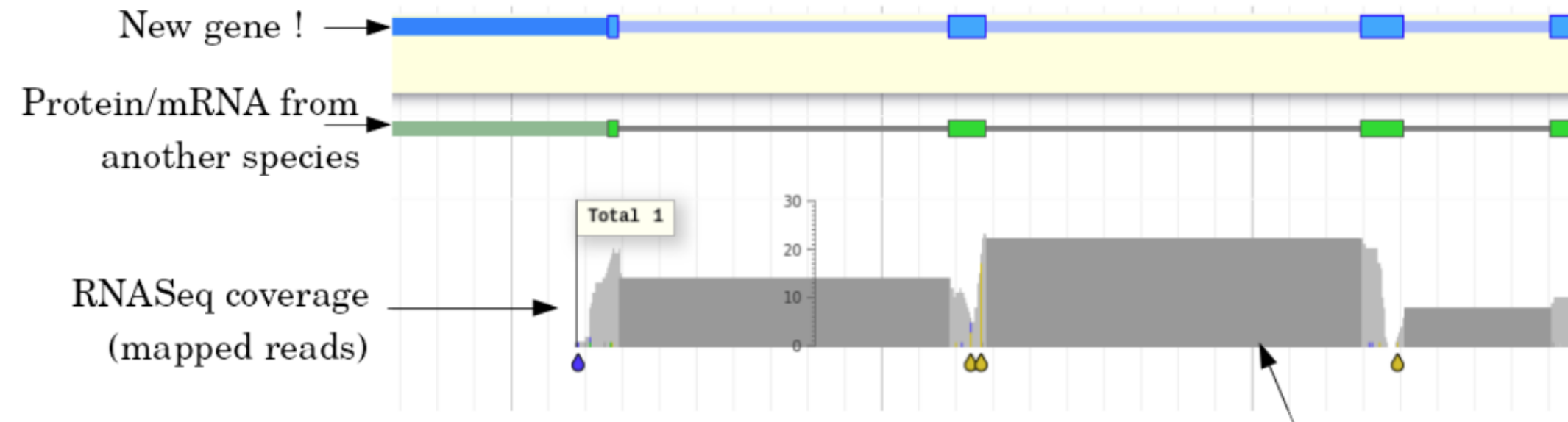


Strategies for identifying coding genes - Structural annotation

Evidence

Multiple pieces of evidence

- Alignment of RNASeq reads
- Alignment of EST or transcripts (same species or closely related species)
- Alignment of proteins (closely related species)



But data unavailable for novel or very distant genes, or unexpressed genes

Strategies for identifying coding genes - Structural annotation

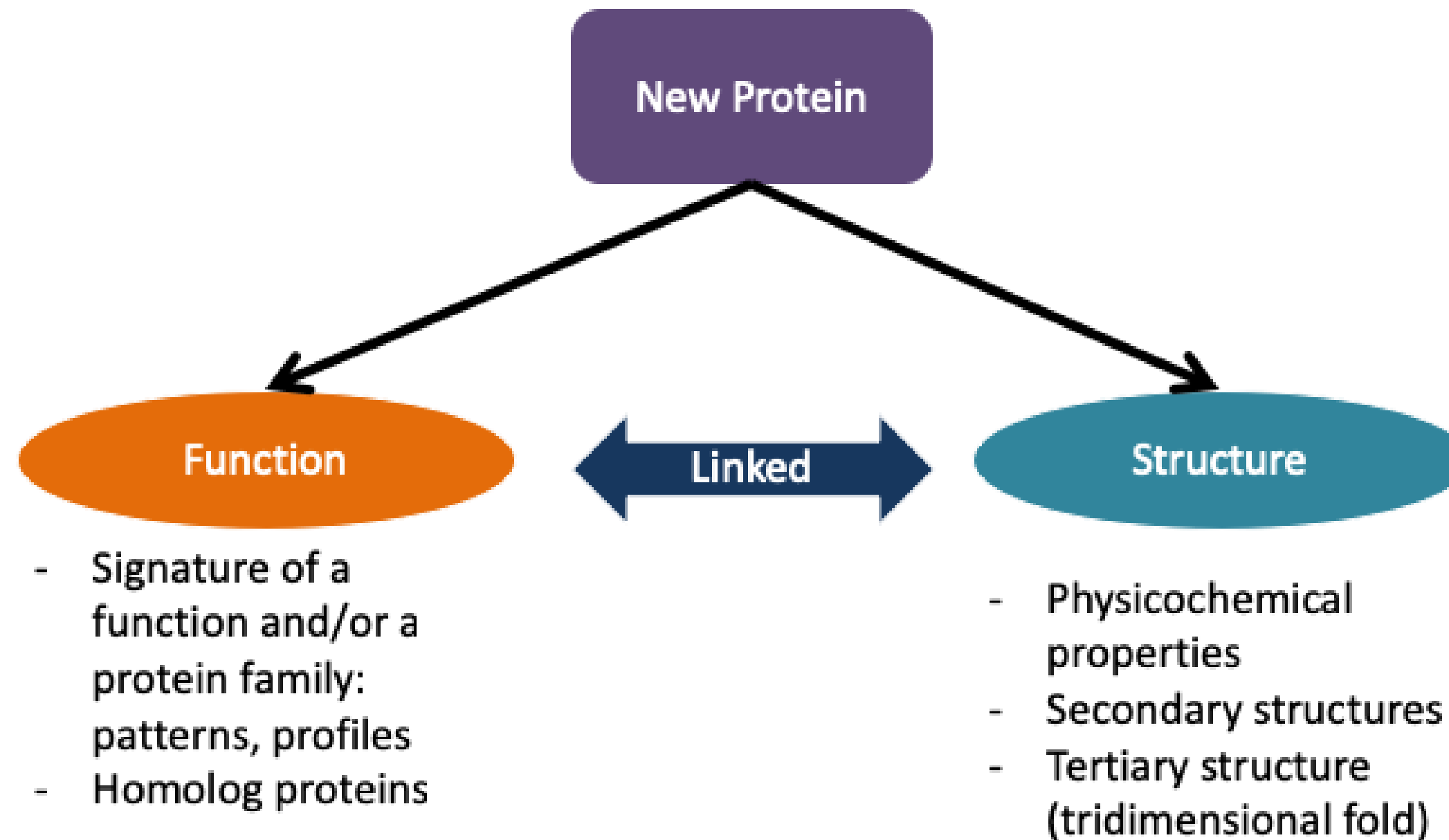
Alignment

- Homology based
- Find similar sequences in DBs
- Might miss proteins that are not in the DB

Ab initio

- Find candidate ORFs
 - Model of RBS
 - Prediction of CDS
- May choose the incorrect start codon
- May miss atypical genes

Functional annotation of the identified proteins



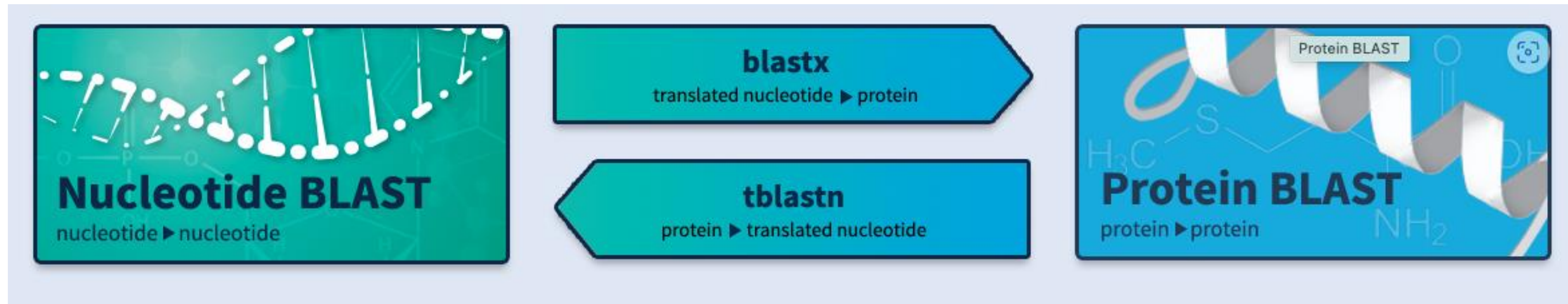
similar sequences => similar structures => similar functions

Functional annotation

Methods

- Similarity search / homology
- Pattern search
- Orthologies
- Comparison against databases:
 - GenBank, NR: sequence databanks
 - InterPro: pattern databank (active sites, protein families, peptide signal ...)
 - EggNOG: databank of orthology relationships + functional annotation

NCBI Blast

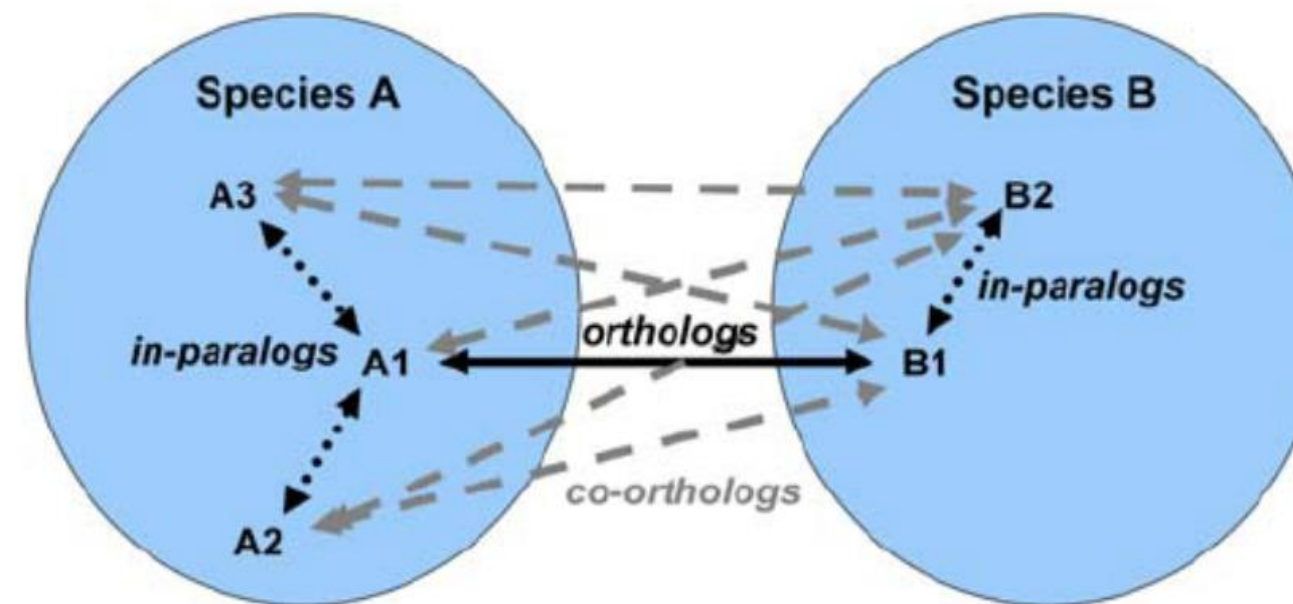


- Blast against **NR**
 - For each protein (or CDS) of the annotation
 - Find the best xx hits
- Huge database, good chances to have a match
- Risk:
 - Spread of "putative xx protein"
 - Spread of low-evidence annotations

Functional annotation

- For each annotated gene
 - Search of orthologous genes in related species
 - Search for paralogues
- Bioinformatics method:
 - Blast all against all transcripts
 - Filtering the best hits
 - Clustering
 - OrthoFinder, OrthoMCL, ...

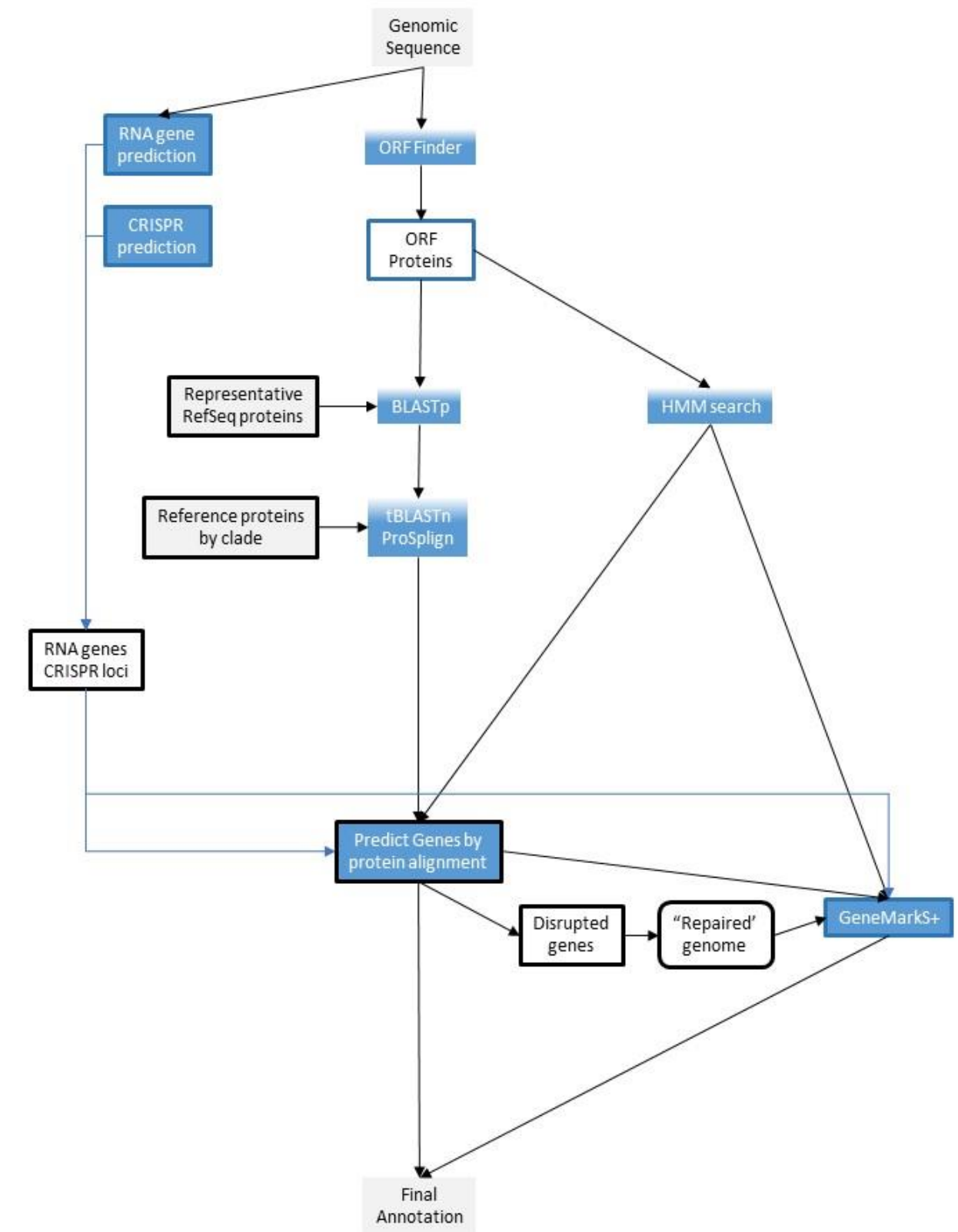
Orthology



Functional annotation

NCBI Prokaryotic Genome Annotation Pipeline

- Structural annotation by comparing ORFs to:
 - libraries of protein models (HMMs),
 - representative RefSeq proteins
 - proteins from well characterized reference genomes.
- GeneMark S+: *ab initio* coding region predictions for regions lacking HMM or protein evidence



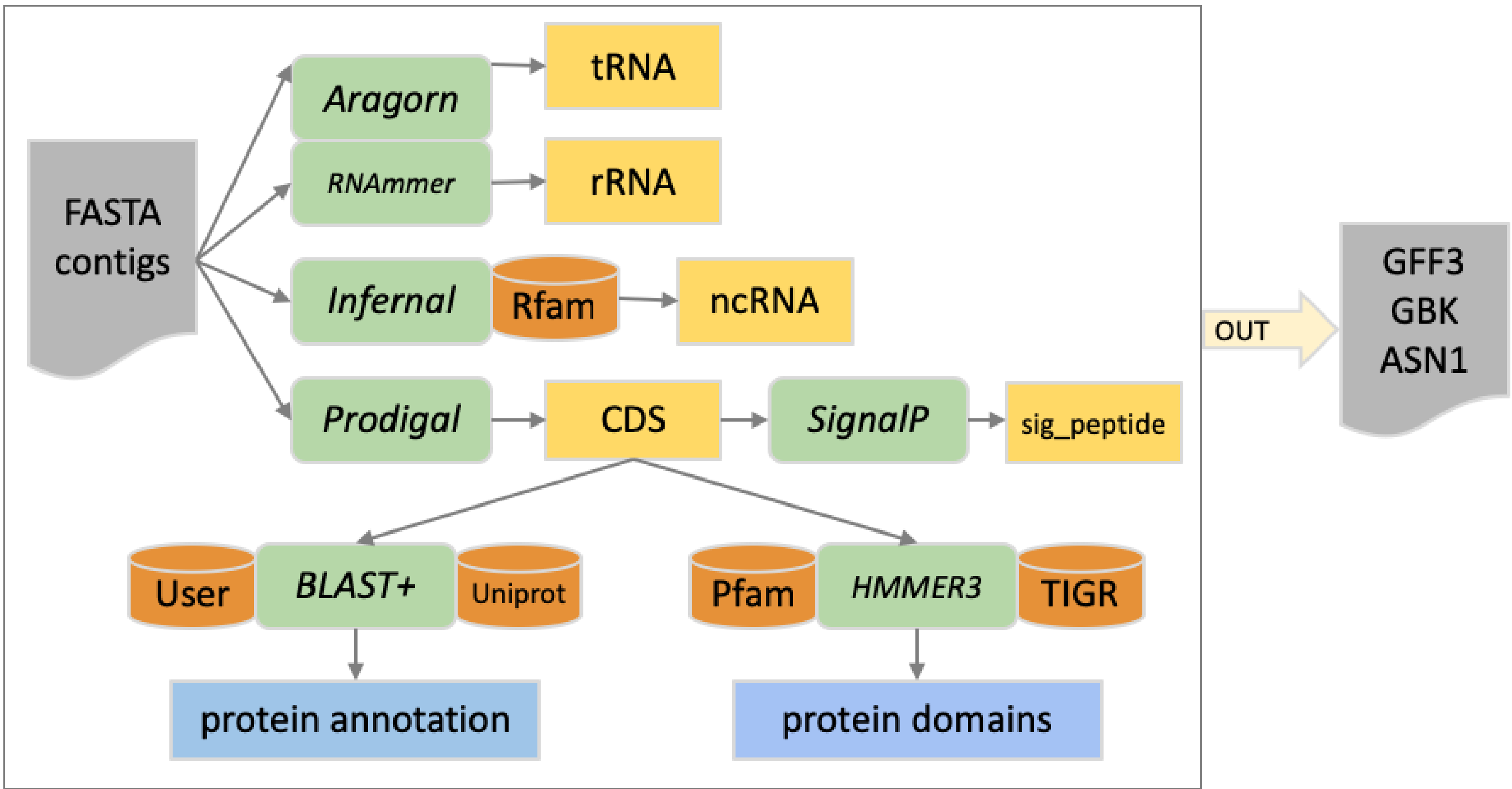
Prokka

- Fast
- Structural and Functional annotation
- Standards compliant
 - GFF3/GBK, TBL/FSA for GenBank
- Archaea, mitochondria, viruses

Feature prediction tools used:

Tool (reference)	Features predicted
Prodigal (Hyatt 2010)	Coding sequence (CDS)
RNAmmer (Lagesen et al. , 2007)	Ribosomal RNA genes (rRNA)
Aragorn (Laslett and Canback, 2004)	Transfer RNA genes
SignalP (Petersen et al. , 2011)	Signal leader peptides
Infernal (Kolbe and Eddy, 2011)	Non-coding RNA





GFF3

GFF3 - The GFF (General Feature Format) format consists of one line per feature, each containing **9** columns of data, plus optional track definition lines.

1.seqid - name of the chromosome or scaffold; chromosome names can be given with or without the 'chr' prefix. Important note: the seq ID must be one used within Ensembl, i.e. a standard chromosome name or an Ensembl identifier such as a scaffold ID, without any additional content such as species or assembly. See the example GFF output below.

2.source - name of the program that generated this feature, or the data source (database or project name)

3.type - type of feature. Must be a term or accession from the SOFA sequence ontology

4.start - Start position of the feature, with sequence numbering starting at 1.

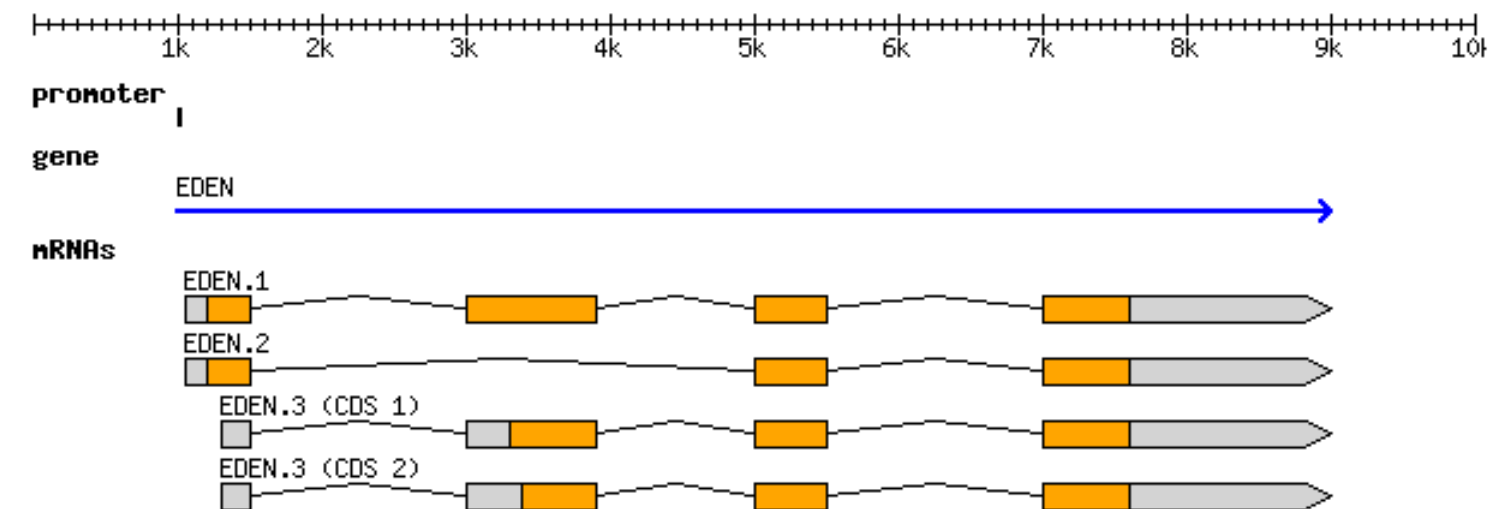
5.end - End position of the feature, with sequence numbering starting at 1.

6.score - A floating point value.

7.strand - defined as + (forward) or - (reverse).

8.phase - One of '0', '1' or '2'. '0' indicates that the first base of the feature is the first base of a codon, '1' that the second base is the first base of a codon, and so on..

attributes - A semicolon-separated list of tag-value pairs, providing additional information about each feature. Some of these tags are predefined, e.g. ID, Name, Alias, Parent - see the GFF documentation for more details



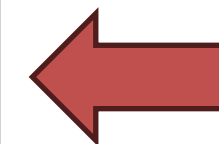
```
0 ##gff-version 3.2.1
1 ##sequence-region ctg123 1 1497228
2 ctg123 . gene 1000 9000 . + . ID=gene00001;Name=EDEN
3 ctg123 . TF_binding_site 1000 1012 . + . ID=tfbs00001;Parent=gene00001
4 ctg123 . mRNA 1050 9000 . + . ID=mRNA00001;Parent=gene00001;Name=EDEN.1
5 ctg123 . mRNA 1050 9000 . + . ID=mRNA00002;Parent=gene00001;Name=EDEN.2
6 ctg123 . mRNA 1300 9000 . + . ID=mRNA00003;Parent=gene00001;Name=EDEN.3
7 ctg123 . exon 1300 1500 . + . ID=exon00001;Parent=mRNA00003
8 ctg123 . exon 1050 1500 . + . ID=exon00002;Parent=mRNA00001,mRNA00002
9 ctg123 . exon 3000 3902 . + . ID=exon00003;Parent=mRNA00001,mRNA00003
10 ctg123 . exon 5000 5500 . + . ID=exon00004;Parent=mRNA00001,mRNA00002,mRNA00003
11 ctg123 . exon 7000 9000 . + . ID=exon00005;Parent=mRNA00001,mRNA00002,mRNA00003
12 ctg123 . CDS 1201 1500 . + 0 ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
13 ctg123 . CDS 3000 3902 . + 0 ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
14 ctg123 . CDS 5000 5500 . + 0 ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
15 ctg123 . CDS 7000 7600 . + 0 ID=cds00001;Parent=mRNA00001;Name=edenprotein.1
16 ctg123 . CDS 1201 1500 . + 0 ID=cds00002;Parent=mRNA00002;Name=edenprotein.2
17 ctg123 . CDS 5000 5500 . + 0 ID=cds00002;Parent=mRNA00002;Name=edenprotein.2
18 ctg123 . CDS 7000 7600 . + 0 ID=cds00002;Parent=mRNA00002;Name=edenprotein.2
19 ctg123 . CDS 3301 3902 . + 0 ID=cds00003;Parent=mRNA00003;Name=edenprotein.3
20 ctg123 . CDS 5000 5500 . + 1 ID=cds00003;Parent=mRNA00003;Name=edenprotein.3
21 ctg123 . CDS 7000 7600 . + 1 ID=cds00003;Parent=mRNA00003;Name=edenprotein.3
22 ctg123 . CDS 3391 3902 . + 0 ID=cds00004;Parent=mRNA00003;Name=edenprotein.4
23 ctg123 . CDS 5000 5500 . + 1 ID=cds00004;Parent=mRNA00003;Name=edenprotein.4
24 ctg123 . CDS 7000 7600 . + 1 ID=cds00004;Parent=mRNA00003;Name=edenprotein.4
```

CS31 - Genômica Computacional

Esse repositório contém materiais de aula para a disciplina de Genômica Computacional da IAMSPE.

Para facilitar execução dos tutoriais, utilizaremos o [Google Cloud Shell](#).

Aula	Data	Tema	Slides	Tutoriais
1	2026-07-27	Introdução e processamento de dados de sequenciamento	Slides	Fastq Quality-Control (QC) tutorial
2	2026-07-28	Montagem de genomas	Slides	Genome Assembly tutorial
3	2026-07-29	Anotacao de genomas	Slides	Genome Annotation tutorial
4	2026-07-30	Mapeando variantes	[Slides]	[Variant mapping tutorial]



https://github.com/RushAlz/IAMSPE-CS31-Genomica_Computacional

Open in Cloud Shell

You are about to clone the repo:

 https://github.com/RushAlz/IAMSPE-CS31-Genomica_Computacional.git

This repo is not officially maintained by Google and is considered untrusted by default.

☒ Trust repo



CANCEL

CONFIRM